SEMESTER 4 • COMMON COURSE • FIELDWORK HANDBOOK

Community Skills in Indian Knowledge System

A syllabus-based bilingual handbook for identifying traditional community knowledge, documenting processes and tools, proposing practical improvements, and applying socially relevant technologies in water, waste, energy, agriculture and irrigation.

COURSE CODE **4001**	SEMESTER **4**
CREDITS **No Credit**	CATEGORY **Common Course**
PERIODS / WEEK **2 (L:2, T:0, P:0)**	PERIODS / SEMESTER **30**
ASSESSMENT **CIA / Record / Fieldwork**	COGNITIVE LEVEL **Applying**

30

Hours of community learning

Observe respectfully, document accurately, improve appropriately and return useful knowledge to the community.

COURSE PURPOSE

What this course expects from students

Appreciate living knowledge

Recognise the skills practised by artisans, workers, farmers and rural communities near the institution.

□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□, □□□□□, □□□□□□□□□□, □□□□□□□□□□□□□ □□□□□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□ □□□□□□□ □□□□□□□□□□□□□.

Connect tradition and technology

Relate traditional processes, tools and implements to the technical knowledge gained in the diploma programme.

Traditional process, tools, implements □□□□□□□□□ diploma-□□□ □□□□□□ technical knowledge-□ □□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Propose useful improvements

Suggest changes that improve safety, quality, productivity, income, comfort and sustainability without destroying traditional value.

□□□□□□□□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□ safety, quality, productivity, income, comfort, sustainability □□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□.

Use socially relevant technologies

Practise or popularise low-cost scientific methods for water, waste and energy management, agriculture and irrigation.

Water, waste, energy management, agriculture, irrigation □□□□□□□□□ community-□□□□ □□□□□□□□□□□□□ low-cost scientific methods □□□□□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□□□.

Document ethically

Obtain consent before photographs, video or personal details and avoid exposing confidential information.

Photo/video □□□□□□□□□□□□□□□□ □□□□□□□ permission □□□□□□□□. □□□□□□□□□□ □□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□□□□.

Return value

Share useful findings, suggestions or awareness material with the participating community whenever possible.

Study □□□□□□□□□□□□□□□□ □□□□□ findings, suggestions, awareness material □□□□□□□ community-□□□□□□□ □□□□□□□□□□□□□□ □□□□□ practice □□□.

SYLLABUS STRUCTURE

Course-outcome and module map

The four modules total 30 hours and lead from observation to documentation, improvement and community application.

CO1 / Module I 10 hours

CO2 / Module II 6 hours

CO3 / Module III 10 hours

CO4 / Module IV 4 hours

CO	Outcome	Duration	Level
CO1	Identify knowledge, skills and practices followed traditionally.	10 hours	Applying

CO	Outcome	Duration	Level
CO 2	Explain processes, methods and implements followed traditionally.	6 hours	Applying
CO 3	Identify improvements in processes and tools to enhance productivity and community living standards.	10 hours	Applying
CO 4	Use socially relevant technologies in water, waste and energy management for the community.	4 hours	Applying

Prerequisites: Art, culture and traditions from secondary-school Social Science, plus programme-relevant technical knowledge acquired up to Semester 3. CO1 and CO2 are strongly mapped to PO5 and PO7, CO3 to PO5, and CO4 to PO5 and PO7.

Editorial clarification: Module IV on page 3 repeats “water” in one sentence. The objectives and CO4 consistently identify the intended areas as **water, waste and energy management**; this lesson follows that interpretation.

CO1 • 10 HOURS

Module I — Identify Traditional Knowledge, Skills and Practices

Conduct a structured local field visit to recognise what the community knows, how the skill is learned, and why the practice remains useful.

Step 1

Choose responsibly

Select a suitable local practice

Choose a skill genuinely practised in the locality, observable safely, and supported by a willing knowledge holder. Examples include pottery, coir work, handloom, bamboo or cane craft, metalwork, carpentry, traditional food processing, indigenous building, farming, seed preservation, irrigation, water harvesting and repair practices.

> □□□□□□□: Locality-□ □□□□□□□□□□□□ □□□□□□□□□□ skill □□□□□□□□□□□□□□□□. □□□□□□ □□□□□□□, □□□□□□□□□□□□□□□□□□□□□□, practitioner □□□□□□□□□□ □□□□□□□□□□□□□.

Step 2

Plan before visiting

Prepare the visit

- ✓ Obtain faculty approval and identify the team.
- ✓ Fix date, time, place and contact person.
- ✓ Prepare questionnaire and observation sheet.
- ✓ Carry notebook, pen, measuring tape and permitted camera.
- ✓ Identify safety needs such as footwear, mask or gloves.
- ✓ Prepare a simple consent statement.

> □□□□□□□: Visit-□□□ □□□□□□□ date, place, contact person, questionnaire, safety, photo permission □□□□□□□ □□□□□□□□□□□□□□.

Step 3

Observe systematically

What to observe

Area	Questions	Evidence
Knowledge source	How was the skill learned—family, apprenticeship or community?	Interview note and learning timeline.
Raw materials	What is used, where is it sourced, and is it seasonal?	Material list and approximate quantity.

Area	Questions	Evidence
Tools	What is each tool called and what does it do?	Labelled sketch or permitted photograph.
Process	What are the steps, sequence, time and quality checks?	Flowchart and time observations.
Skill judgement	Which steps rely on touch, sound, colour or experience?	Practitioner explanation and student interpretation.
Safety	Are there heat, dust, sharp tools, heavy loads or awkward posture?	Hazard list without blaming the worker.
Economics	What affects cost, demand, output and income?	Only voluntarily shared approximate data.
Culture/environment	What local identity or ecological value does the practice carry?	Context note and examples.

□□□□□□: Materials, tools, process sequence, skill judgement, safety, cost, cultural value, environmental impact □□□□□□ □□□□□□□□□□□ □□□□□□□□□□□□□□.

Field questionnaire — ready to use

1. Name of the skill/practice:
2. Name of artisan/community group (only with consent):
3. Location and date of visit:
4. How long has this practice existed in the family/community?
5. How did the practitioner learn the skill?
6. What raw materials are used and where are they sourced?
7. What tools and implements are used? State each function.
8. What are the main process steps?
9. Which step requires the greatest experience or judgement?
10. What quality checks are followed?
11. What difficulties are faced—time, cost, safety, demand, material or transport?
12. Have tools or methods changed in recent years?
13. What improvement would the practitioner welcome?
14. What should not be changed because it is essential to quality or tradition?
15. May photographs/video be used in the academic record? Yes / No / Limited.

CO1 evidence: approved field-visit plan, completed questionnaire, observation notes, labelled evidence and identification of the traditional knowledge embedded in the practice.

CO2 • 6 HOURS

Module II — Explain the Process, Methods and Implements

Convert raw field observations into a clear report supported by appropriate photographs, sketches, tables and videos.

Report Structure

6 hours • Applying

Recommended report order

1. **Cover page:** course, title, student details, institution and academic year.
2. **Certificate:** bonafide statement signed as required.
3. **Acknowledgement:** thank practitioner, community, guide and institution.
4. **Abstract:** 150–250 words covering skill, method, finding and improvement.
5. **Introduction:** locality, background, relevance and objectives.
6. **Methodology:** visit, observation, interview, questionnaire and documentation.
7. **Traditional process:** materials, tools, steps, skills and quality checks.
8. **Findings:** strengths, limitations, safety, productivity and environment.
9. **Suggestions:** technically justified, low-cost and acceptable improvements.
10. **Conclusion and annexures:** learning, questionnaire, diary, evidence and consent.

□□□□□□□: Report □□□ □□□□□□□ □□□□; Introduction → Methodology → Process → Findings → Improvements → Conclusion □□□□ technical order □□□□□□□□□□.

Photo Evidence

Use photographs correctly

- ✓ Use only permitted images.
- ✓ Show process, tool or result—not unnecessary personal details.
- ✓ Add figure number, caption, location and date.

- ✓ Refer to every important figure in the text.
- ✓ Do not stretch, blur or over-edit evidence.

Caption: “Figure 3. Manual shaping stage using a locally fabricated wooden tool.”

Photo-□□□□□ figure number, caption, context □□□□□□. Permission □□□□□□□□□ face/photo □□□□□□□□□□□.

Video

Use video in presentation

- ✓ Keep clips short and relevant.
- ✓ Introduce the process before playing.
- ✓ Remove private conversation.
- ✓ Add subtitles where sound is unclear.
- ✓ Explain the technical point shown.

Video □□□□□□□ □□□□□□□□□□□□□□□ explanation-□□ □□□□□□□□. “□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□?” □□□□□□ □□□□□□.

Process Map

Convert observation into an engineering record

For each step, record input, tool, action, approximate time, skill judgement, quality check, output and hazard.

□□□ process step-□□□ input, tool, action, time, skill, quality check, output, hazard □□□□□□□ □□□□□□□.

CO2 evidence: complete process flow, tool table, labelled documentary evidence, clear report writing and a short presentation.

CO3 • 10 HOURS

Module III — Improve Processes and Tools

Identify improvements that are technically sound, affordable, safe, respectful and acceptable to the community.

Improvement Cycle

10 hours • Applying

From observation to recommendation

1. **Define the problem:** state what happens, where, how often and who is affected.
2. **Protect traditional value:** identify what must not be lost—quality, identity, skill or cultural meaning.
3. **Find root causes:** check material, method, tool, person, environment and measurement.
4. **Generate options:** produce more than one solution, including a low-cost option.
5. **Evaluate feasibility:** cost, safety, maintenance, energy, training, local availability, acceptance and benefit.
6. **Prototype:** test a small change before recommending full adoption.
7. **Measure impact:** compare time, output, defects, effort, safety or resource use.

□□□□□□□: Problem □□□□ □□□ expensive machine suggest □□□□□□□□□□. Root cause □□□□□□□□□□ low-cost, maintainable, locally acceptable option □□□□□□□□□□□□□□□.

Improvement Areas

What can be improved?

Area	Possible improvement	Evidence
Safety	Guard, stable surface, ventilation, dust control, lighting, safer handle or PPE awareness.	Hazard observation and risk reduction.
Ergonomics	Correct work height, seating, tool grip and reduced bending.	Posture comparison and user feedback.
Productivity	Layout, jig, fixture, template, batch organisation or reduced waiting.	Time study or output comparison.
Quality	Gauge, size guide, moisture check, temperature indicator or checklist.	Defect/rework comparison.

Area	Possible improvement	Evidence
Material use	Accurate cutting, off-cut reuse, storage or controlled mixing.	Waste estimate or material balance.
Energy	Insulation, improved airflow, efficient motor, solar drying or daylight use.	Fuel/electricity estimate and safety check.
Marketing	Labelling, packaging, digital catalogue and direct customer link.	Community interest and practical plan.

Feasibility

Score each proposal

1–5 Cost	**1–5** Safety
1–5 Benefit	**1–5** Maintenance
1–5 Acceptance	

Explain each score and select the option with the best practical balance—not merely the highest technology.

Technology advanced □□□□□□□□□ □□□□□□□ criterion □□□□. Cost, benefit, maintenance, acceptance □□□□□□□ □□□□□□□□.

Recommendation

Write recommendations correctly

- ✓ State the observed issue.
- ✓ Give the technical reason.
- ✓ Describe the proposed modification.
- ✓ Estimate materials, cost and maintenance.
- ✓ State expected measurable benefit.
- ✓ Mention limitations and training.
- ✓ Record practitioner response.

“Improve □□□□□□□□” □□□□□□ □□□□□□□ □□□□□□□□□. □□□□□□ change, □□□□□□□, □□□□ cost, □□□□□□ benefit □□□□□□ □□□□□□□□□□□□□□□.

CO3 evidence: detailed case study, present-condition analysis, at least two alternatives, feasibility comparison, final recommendation and expected impact.

CO4 • 4 HOURS

Module IV — Socially Relevant Technology

Practise, popularise or improve scientific methods in water, waste and energy management, agriculture and irrigation, supported by documentary evidence.

Water

Water management

- ✓ Leak detection and repair awareness.
- ✓ Rainwater collection and first-flush awareness.
- ✓ Safe storage and covered containers.
- ✓ Simple water-use audit.
- ✓ Recharge and runoff control appropriate to the site.

Water project-□ local conditions, water quality, safety, maintenance □□□□□□□ □□□□□□□□□□□□□□.

Waste

Waste management

- ✓ Source segregation into appropriate categories.
- ✓ Composting suitable biodegradable waste.
- ✓ Safe collection of sharp, chemical or electronic waste.
- ✓ Material reuse and reduction at source.
- ✓ Awareness signage and monitoring record.

Hazardous/e-waste □□□□□□□ compost/reuse stream-□ □□□□□□□□□□□□.

Energy

Non-conventional and efficient energy

- ✓ Solar drying or lighting demonstration.
- ✓ Improved daylight and ventilation.
- ✓ Efficient cooking or process heat.
- ✓ Energy-use survey and awareness.
- ✓ Biogas only with expert guidance.

Electrical/gas work expert supervision □□□□□□□□ □□□□□□□□□□.

Agriculture

Agriculture and irrigation

- ✓ Mulching and moisture conservation.
- ✓ Drip or controlled irrigation demonstration.
- ✓ Simple soil-moisture observation.
- ✓ Seed and produce storage improvement.
- ✓ Tool ergonomics and safe handling.

Farmer experience □□□□□□□□□□; suggestion field condition-□□□□ □□□□□□□□□□□□□□□□□□□□□□□□ verify □□□□□□□□.

Project Plan

Four-hour community activity format

1. **Need:** select one real and manageable issue with the community.
2. **Baseline:** record present condition using count, measurement, photograph or survey.
3. **Intervention:** demonstrate, install, practise or popularise the selected method safely.
4. **Awareness:** provide a bilingual poster, checklist or demonstration.
5. **Verification:** record participation, feedback and measurable result.
6. **Documentation:** include before/after evidence, limitations and follow-up.

Activity □□□□□□□□□□□□□□ real community benefit □□□□□□□□□□. Before/after evidence, feedback, limitations □□□□□□□□□□□□□□□□.

CO4 evidence: need statement, baseline, safe intervention, awareness material, documentary evidence, feedback and follow-up recommendation.

Complete record and submission structure

The syllabus states that CIA is to be followed as included in the handbook. This section organises the evidence expected from a field-oriented course.

Before fieldwork

- ✓ Topic approval
- ✓ Team and roles
- ✓ Visit plan
- ✓ Questionnaire
- ✓ Safety and consent plan

During fieldwork

- ✓ Dated field diary
- ✓ Interview notes
- ✓ Process timing
- ✓ Tool sketches
- ✓ Permitted photo/video

After fieldwork

- ✓ Process flowchart
- ✓ Analysis tables
- ✓ Improvement matrix
- ✓ Final report
- ✓ Presentation and viva

Bonafide report contents page

Possible CIA criterion	Quality evidence
Participation	Prepared visit, respectful interaction, observation and individual contribution.
Technical understanding	Correct process, labelled tools, flowchart and programme connection.
Documentation	Clear record, captions, dates, questionnaire and ethical consent.
Analysis	Root-cause reasoning, alternatives, feasibility and measurable benefit.
Community relevance	Practical, safe, affordable and acceptable recommendation.
Presentation	Clear explanation, team coordination and honest limitations.

Do not fabricate evidence: A smaller genuine study with real dates, observations and limitations is better than a large report containing invented interviews, costs, photographs or results.

COMPLETE MODEL CASE STUDY

Hypothetical sample — Traditional pottery process improvement

This model demonstrates report structure and reasoning. Students must replace every detail with verified field observations.

Model report — compact complete version

TITLE
Study of a Traditional Pottery Process and Proposal for a Low-Cost Ergonomic and Drying Improvement

ABSTRACT
This hypothetical study documents clay preparation, wedging, wheel forming, initial drying, finishing, firing and inspection. It focuses on knowledge used to judge clay consistency, control hand pressure and identify drying condition. Two possible problems are analysed: repeated bending during clay preparation and cracking caused by uneven drying. A raised work platform and shaded ventilated drying rack are proposed. The options are evaluated for cost, safety, maintenance, local material availability and acceptance. Actual student reports must use verified field data.

OBJECTIVES
1. Identify traditional knowledge and tools used in pottery.
2. Explain the process and skill-dependent stages.
3. Identify safety, ergonomic and productivity issues.
4. Develop low-cost improvement proposals.
5. Prepare evidence and recommendations.

METHODOLOGY
A real project should use an approved visit, practitioner interview, observation, time notes, tool sketches, permitted photographs and feedback discussion. Personal or commercial information must be recorded only with consent.

PROCESS FLOW
Clay selection → stone removal → water addition → kneading/wedging → wheel forming → initial drying → finishing → final drying → firing → inspection.

TRADITIONAL KNOWLEDGE
- Judging clay moisture by touch and behaviour.
- Coordinating wheel speed and hand pressure.
- Recognising drying condition from colour and surface feel.
- Arranging products during firing through experience.
- Repairing small defects before complete drying.

TOOLS
Potter's wheel, wooden shaping tools, cutting wire, water container, finishing tool, drying surface and kiln/firing arrangement.

PROBLEM 1 — ERGONOMIC STRAIN
Condition to verify: low-level clay preparation may require repeated bending.
Root causes: low work height, poor material placement and manual handling.
Alternative A: raised stable preparation platform using local material.
Alternative B: low-cost seating and repositioned material containers.
Recommendation: stable raised platform after confirming preferred height.
Expected benefit: reduced bending and easier handling.
Limitation: it must not obstruct the traditional method and must remain stable when wet.

PROBLEM 2 — UNEVEN DRYING
Condition to verify: direct sun or uneven airflow may cause non-uniform drying and cracking.
Root causes: rapid moisture loss, wet-ground contact and insufficient protected space.
Alternative A: shaded ventilated rack with multiple levels.
Alternative B: movable light cover and regular turning schedule.
Recommendation: locally fabricated shaded rack with a drying checklist.
Expected benefit: more uniform drying, cleaner storage and reduced handling damage.

FEASIBILITY
Raised platform — Cost 4/5, Safety 4/5, Benefit 4/5, Maintenance 5/5, Acceptance to be confirmed.
Drying rack — Cost 4/5, Safety 4/5, Benefit 5/5, Maintenance 4/5, Acceptance to be confirmed.

SOCIALLY RELEVANT TECHNOLOGY LINK
A rainwater collection container for non-potable process use may be studied only after checking roof cleanliness, first-flush arrangement, safe storage and local suitability.

COMMUNITY FEEDBACK
A real report must record the practitioner’s response: useful / not useful / needs modification, preferred dimensions and material, cost concern and willingness for a small trial.

CONCLUSION
Traditional pottery contains significant material, process and quality knowledge. Engineering improvement should support the practitioner rather than replace valuable judgement. Suitable recommendations are low-cost, safe, repairable and accepted by the user.

EVIDENCE TO ATTACH
Field plan, questionnaire, tool sketch, process flowchart, permitted photographs, measurements, feasibility matrix, practitioner feedback and team reflection.

□□□□□□□: Model report copy □□□□□□□□□□. □□□□□□□□ field visit data, actual photos, practitioner feedback, measurements □□□□□□□ □□□□□□□□□□□□□.

PRACTICE VIVA / WRITTEN REVIEW

Questions with model answers

This is a CIA and viva practice set; the syllabus identifies the subject as a fieldwork-oriented no-credit course.

1. What is traditional knowledge?

It is knowledge, skill, judgement and practice developed and transmitted within a community over time, including material selection, tool use, process sequence, quality judgement and cultural meaning.

> Community-□□□ □□□□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□ □□□□□ □□□□□, skill, judgement, practice □□□□□□□□□□ traditional knowledge.

2. Why is consent necessary?

Consent respects privacy, rights and ownership of the knowledge holder and clarifies whether names, photographs, videos and process details may be used academically.

3. Observation and interview evidence—what is the difference?

Observation is what the student directly sees, measures or records. Interview evidence is the practitioner's explanation, experience or opinion. A report should distinguish them.

4. What is appropriate technology?

A safe, affordable, locally maintainable, resource-conscious, culturally acceptable and useful improvement under actual working conditions.

5. How can productivity be measured?

By time per unit, units per day, waiting time, material loss, defects, rework, effort or energy use, depending on the process.

6. What belongs in a process flowchart?

Inputs, preparation, core processing, finishing, output, inspections, waiting stages, rework and relevant waste streams.

7. Give two socially relevant technology examples.

A water-use audit with leak-reduction awareness; source segregation and composting of suitable biodegradable waste; solar drying; controlled-irrigation demonstration.

8. Why not replace traditional methods immediately?

A method may contain hidden quality, cultural, environmental or market value. Replacement without participation can reduce identity or create maintenance and cost problems.

9. What documentary evidence is expected?

Dated notes, questionnaire, sketches, permitted photographs/video, measurements, process flow, consent, calculations, feasibility table and community feedback.

10. State the core learning of the course.

Technical education becomes socially useful when students understand local knowledge respectfully, analyse it systematically and develop improvements with the community.

> Local knowledge □□□□□□□ □□□□□□□ community-□□□□□□ □□□□□□□ practical improvement □□□□□□□□□□□□ technical education □□□□□□□□□□□ □□□□□□□□□□□□□□□□□□.

REFERENCES FROM THE SYLLABUS

Text and reference material

Code	Book / Source
T1	Nirmal Sengupta, *Traditional Knowledge in Modern India*, Springer.
R2	Kerala State Planning Board, *Economic Review*, relevant year.
R3	M. Lalithambika, *Techno-socio-economic survey on the living and working conditions of the traditional potter communities of Kerala*, IRTC Publications.
R4	A. Sreedhara Menon, *Legacy of Kerala*, DC Books.